InView - Cloud SCADA

INIT Technologies

Who We Are

INDAS Group / INIT Technologies — Novi Sad, Serbia

- INDAS Group founded **1992** — 30+ years in industrial software
- INIT Technologies spun off **2020** — focused on software product R&D
- **inVIEW** first commercial release: **2011**, microservices + cloud-native since 2017
- Full-stack capability: field protocols → edge → cloud → AI/ML → web/mobile
- In-house R&D — no outsourcing, continuous development

Introducing inVIEW

Cloud-driven. Edge automation. AI-enhanced.

inVIEW is an **Industrial Cloud SCADA** that connects your machines, devices, databases and software — and lets you build real-time dashboards, SCADA visualizations, and MES workflows.

Three core promises:

- **See everything** — real-time data from every device, anywhere
- **Control everything** — remote commands, automation, alarms
- **Understand everything** — AI/ML that finds patterns you can't see manually

> *"It's not just monitoring. It's an evolution of how you run your production."*

THE PLATFORM — 6 MODULES

CORE ENGINE — Real-Time SCADA at the Center

The heart of the platform. Proven to scale solutoin

- Real world data inside the Platform - live variable streaming
- **Remote control** of field devices — write commands from anywhere in the world
- **Alarm management** — configurable thresholds, deadband, double-check validation, false alarm prevention
- Multi-source data aggregation — heterogeneous devices, one unified view
- Multi-channel notifications: SMS, Email, Push

#Proven to Scale

- **100,000 simultaneous screen views** — live, on microservices architecture
- Multi-tenant: multiple enterprise customers on shared infrastructure
- Kubernetes orchestration — horizontal scaling, zero-downtime deployments

DEVICES — Connect Anything

We speak the language of your machines.

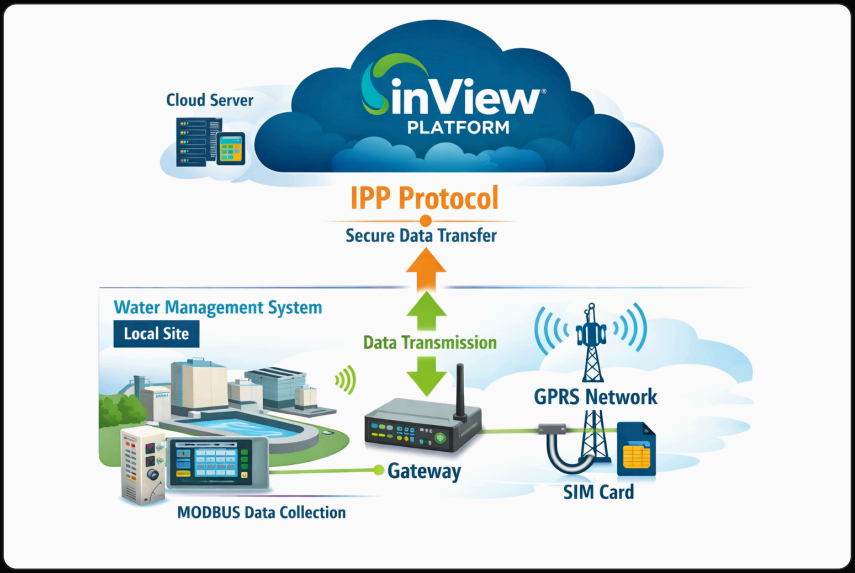
Natively supported protocols:

- **OPC UA / OPC DA** — industrial interoperability standard
- **IEC 60870-5-104** — power grid, utilities, telecontrol
- **MQTT / LoRaWAN** — modern IoT sensors
- **Modbus TCP** — most common industrial protocol
- **Siemens S7** — direct PLC connection
- **i3x** — inVIEW proprietary protocol for gateway communication

inVIEW IoT Gateways — for everything else:

- Hardware gateways that bridge legacy equipment to the platform
- Support: **Modbus RTU/TCP, Siemens S7, Ethernet/IP, Profinet, OPC**
- **Custom script execution on the gateway** — JavaScript logic runs locally, at the source
- **Store & Forward** — buffers data locally during connectivity loss, syncs automatically
- Gateway software is open for custom driver development

> *"If your machine generates data, we can connect to it — and process it right where it lives."*



NO-CODE IDE — Your Team Builds It

The people who know your process — build the application.

- **Drag-and-drop screen builder** — no programming required
- Configurator runs entirely in a browser — no software to install
- Your technician builds a new screen, not your software developer

One tool — web and mobile:

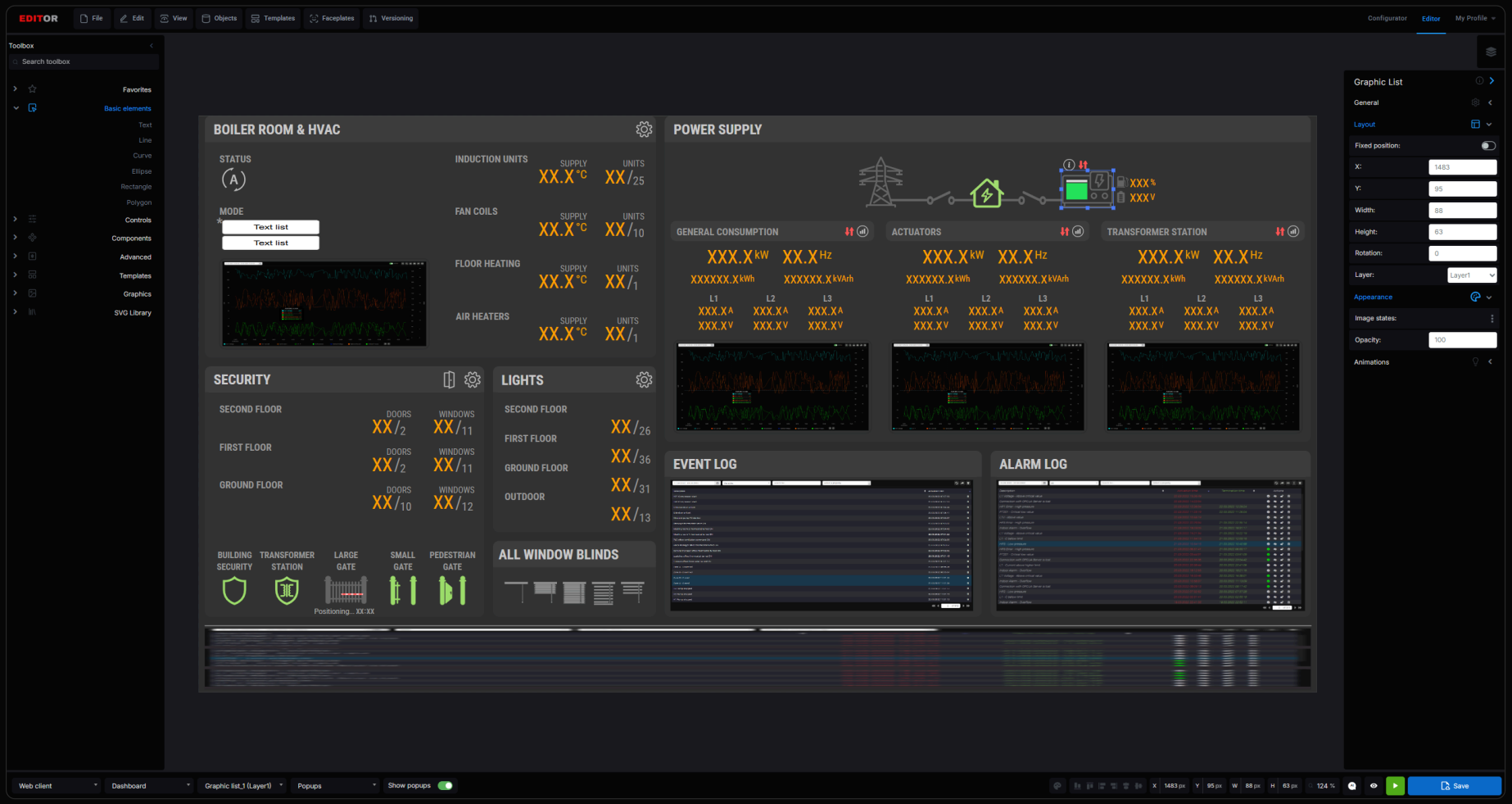
- Screens built in the IDE render on **web browser** and in the **native mobile app** (iOS & Android)
- No separate mobile development — same screen definition, same data, same configuration

What this means in practice:

- Not dependent on a vendor for every configuration change
- IT team or production engineers own the setup
- **Templates** — configure one machine, replicate to 50 identical ones instantly
- New production line → new screen in days, not months

> *"Most SCADA vendors charge you every time you want to change a label. Your team does it themselves — and it works on every device."*

NO-CODE IDE — YOUR TEAM BUILDS IT



NO-CODE IDE — YOUR TEAM BUILDS IT

OUTDOOR 11.9 °C

20.03.2026 08:29
BMS Nova Zgrada

BOILER ROOM & HVAC

STATUS

MODE
*HEATING
(GAS BOILERS)

	SUPPLY	UNITS
INDUCTION UNITS	33.3 °C	16/25
FAN COILS	32.1 °C	4/10
FLOOR HEATING	22.2 °C	0/1
AIR HEATERS	48.9 °C	1/1

POWER SUPPLY

95 %
13.4 V

GENERAL CONSUMPTION	ACTUATORS	TRANSFORMER STATION
29.2 kW	5.8 kW	29.3 kW
50.0 Hz	50.0 Hz	50.0 Hz
129600.3 kWh	62322.6 kWh	145561.0 kWh
4303.1 kVArh	2411.9 kVArh	7322.4 kVArh
L1 48.6 A 234.7 V	L1 11.1 A 234.7 V	L1 32.6 A 235.4 V
L2 53.4 A 235.6 V	L2 10.9 A 235.4 V	L2 48.3 A 235.5 V
L3 32.9 A 234.9 V	L3 12.5 A 234.6 V	L3 53.8 A 236.1 V

SECURITY

	DOORS	WINDOWS
SECOND FLOOR	0/2	0/11
FIRST FLOOR	1/2	0/11
GROUND FLOOR	0/10	0/12

BUILDING SECURITY | TRANSFORMER STATION | LARGE GATE | SMALL GATE | PEDESTRIAN GATE

LIGHTS

SECOND FLOOR	4/26
FIRST FLOOR	11/36
GROUND FLOOR	16/31
OUTDOOR	0/13

ALL WINDOW BLINDS

EVENT LOG

Pedestrian gate closed	2026-03-20 08:21:15
Pedestrian gate opened	2026-03-20 08:21:11
Pedestrian gate closed	2026-03-20 08:15:41
Pedestrian gate opened	2026-03-20 08:15:37
Retail door unlocked	2026-03-20 08:00:00

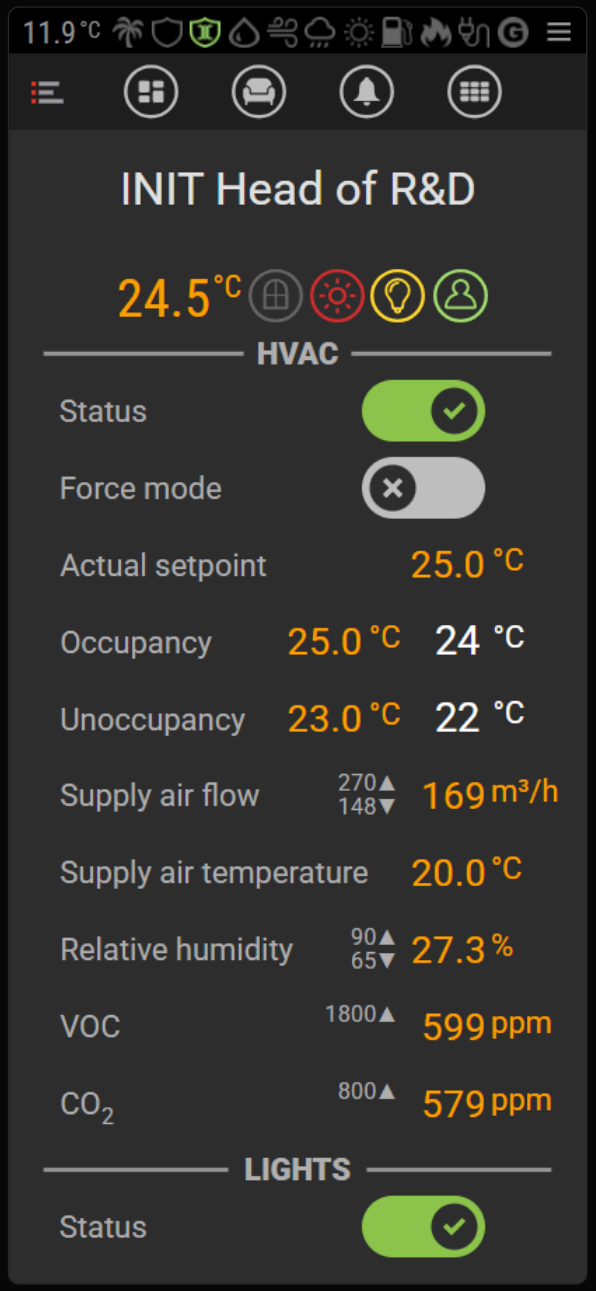
ALARM LOG

Building entrance door is opened for too long	2026-03-20 07:53:48	2026-(
Gateway BMS PLC losing connection to broker	2026-03-19 23:51:16	2026-(
Gateway BMS PLC losing connection to broker	2026-03-19 22:31:06	2026-(
Gateway BMS PLC losing connection to broker	2026-03-19 22:23:16	2026-(
Building entrance door is opened for too long	2026-03-19 16:02:03	2026-(

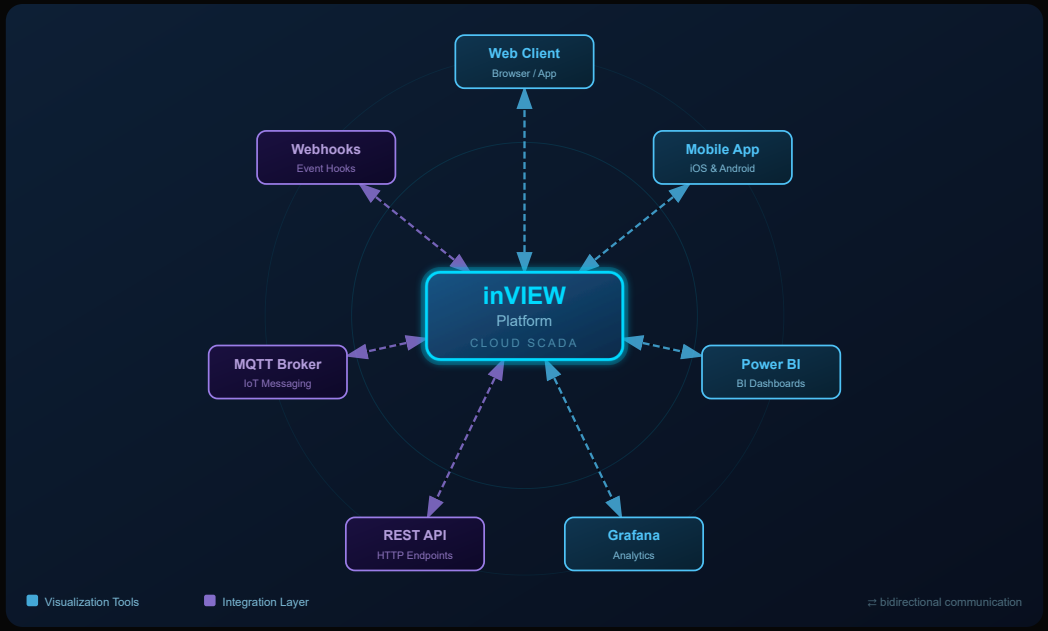
Gutter heater has error 2026-02-14 14:42:11

NO-CODE IDE — YOUR TEAM BUILDS IT

NO-CODE IDE — YOUR TEAM BUILDS IT



ENDPOINTS — Connect to Everything Outside



inVIEW doesn't live in isolation. It talks to your entire ecosystem.

InView visualisation tools

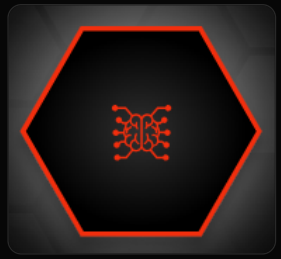
- **Web Client** — expose live and historical tag data via standard InView Client — any browser, app
- **Mobile App** — native iOS & Android app — same screens built in the No-Code IDE, always in sync, works on the shop floor
- **Power BI** — connect inVIEW as a data source for business-level reporting, KPI dashboards, and management overviews
- **Grafana** — plug inVIEW into Grafana for open-source analytical dashboards and time-series visualization

External systems to integrate with via

- **REST API** — standard HTTP endpoints for any external system
- **MQTT broker** — publish data to any MQTT subscriber
- **Webhooks / Event Hooks** — trigger external actions on alarms or data events

> *"Any system that can send or receive HTTP, WebSocket, or MQTT can integrate with inVIEW."*

AI / ML — Your System Gets Smarter Over Time



Built in. Not bolted on.

What the AI layer does:

- **Data analytics AI** - Anomaly Detection, **Self-Prediction**
- **RealTime Calculations** - In realtime have the AI output of your data
- **AI control** - take actions from AI directly
- **Edge RealTime AI Control** - download AI calculations to the Edge

RealTime AI calculaton output > *"Your press temperature starts drifting at 02:00 AM. By 06:00 AM it would have caused a quality defect. The system flags it at 02:15 AM. Your night shift fixes it before morning. That's pattern recognition on your own historical data."*

EDGE — Intelligence Where the Data Is Born

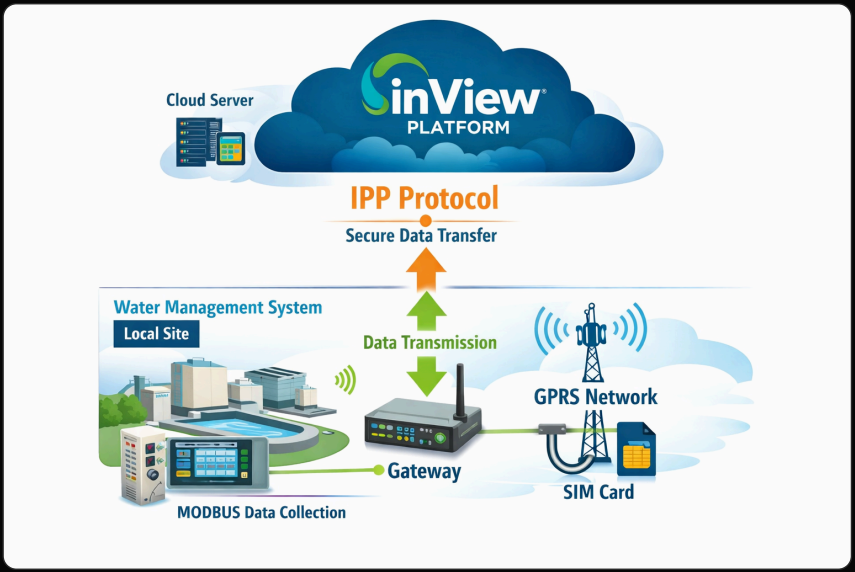
Processing logic runs directly on the edge device — at the machine, at the site.

What edge computing means here:

- Automation scripts run **locally** — no cloud round-trip for time-sensitive decisions
- Works fully **offline** — no internet connection required for local operations
- Reduces bandwidth — only relevant, processed data goes upstream
- Latency-sensitive control stays on-site

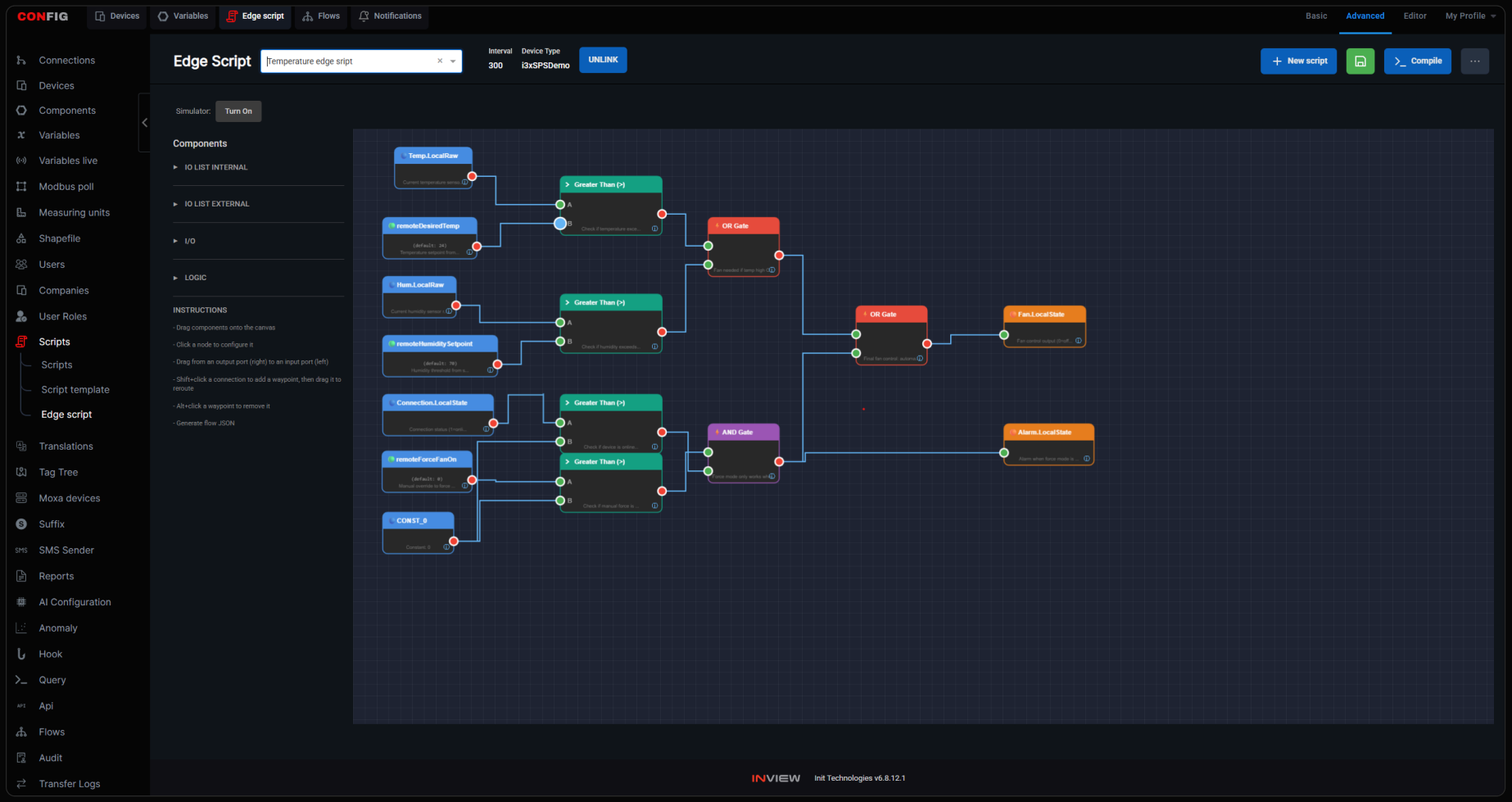
Edge + Gateway combination:

- inVIEW Gateway runs edge scripts at the protocol bridge level
- Custom JavaScript logic executes at the source
- Local alarm detection, filtering, and aggregation before cloud sync
- Scales to hundreds of distributed edge nodes from one central management interface



> *"You have a press line with poor connectivity. With Edge, that line is fully autonomous — it monitors, reacts, logs — with or without a live cloud connection."*

EDGE SCRIPTS EDITOR



EDGE SCRIPTS EDITOR

CONFIG | Devices | Variables | Edge script | Flows | Notifications | Basic | Advanced | Editor | My Profile

- Connections
- Devices
- Components
- Variables
- Variables live
- Modbus poll
- Measuring units
- Shapefile
- Users
- Companies
- User Roles
- Scripts
  - Scripts
  - Script template
  - Edge script
- Translations
- Tag Tree
- Moxa devices
- Suffix
- SMS Sender
- Reports
- AI Configuration
- Anomaly
- Hook
- Query
- Api

Edge Script

Bottles Edge Script

Interval: 10

Device Type: i3xSPSDemo

UNLINK

New script | Compile

```
double inBottles = live.readVariableValueBySufix("Trends.InBottlesCount");
double outnBottles = live.readVariableValueBySufix("Trends.OutBottlesAClass");

live.writeVariableValueBySufix("Trends.OutBottlesScrap", inBottles-outnBottles);


/*live.writeVariableValueBySufix("EdgeInternal.BottleFilling1PreviousHourState", live.readVariableValueBySufix("Trends.BottleFilling1"));
live.writeVariableValueBySufix("EdgeInternal.BottleFilling2PreviousHourState", live.readVariableValueBySufix("Trends.BottleFilling2"));
live.writeVariableValueBySufix("EdgeInternal.BottleFilling3PreviousHourState", live.readVariableValueBySufix("Trends.BottleFilling3"));
*/

int fil1 = (int)(System.currentTimeMillis()%100);
int fil2 = fil1 * 2;
int fil3 = fil1 + 20;

live.writeVariableValueBySufix("Trends.BottleFilling1", fil1);
live.writeVariableValueBySufix("Trends.BottleFilling2", fil2);
live.writeVariableValueBySufix("Trends.BottleFilling3", fil3);

int fillLimitMaint = 10;


boolean f1M = isReadyToMaintenance(fil1);
boolean f2M = isReadyToMaintenance(fil2);
boolean f3M = isReadyToMaintenance(fil3);

live.writeVariableValueBySufix("Trends.BottleFilling1Maintenance", f1M ? 1:0);
live.writeVariableValueBySufix("Trends.BottleFilling2Maintenance", f2M ? 1:0);
live.writeVariableValueBySufix("Trends.BottleFilling3Maintenance", f3M ? 1:0);

// live.writeVariableValueBySufix("EdgeInternal.ScriptExecStat", "Done script!");
```

Additional methods

```
static int FILL_LIMIT_MAINTENANCE = 10;

boolean isReadyToMaintenance(int fil){
    return fil < FILL_LIMIT_MAINTENANCE;
}
```

EDGE SCRIPTS EDITOR

CONFIG | Devices | Variables | Edge script | Flows | Notifications | Basic | Advanced | Editor | My Profile

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- Hook
- Query
- Api
- Flows
- Audit
- Transfer Logs
- Project settings

Menu > Flow > Diagram

Flow

Project | Create New

Search Flow (by Connection & IMEI) | New Flow

	Flow Name	Author	Connection	IP Address	IMEI	Architecture	Status	
Offline	testst	jfrkdlja	-	-	-	-	-	Go to Diagram
Offline	NEW FLOW 6.3.2025.	sadasd	-	-	-	-	-	Go to Diagram
Offline	mio	yo	-	-	-	-	-	Go to Diagram
Offline	sbTestFlow	sb	-	-	-	-	-	Go to Diagram
Offline	Test OR	TEST	-	-	-	-	-	Go to Diagram
Online	Skid P-205 Main Pump Control	zile	Platform Executor	nodered-service:1880	PlatformExecutor	DETERMINISM	RUNNING	Go to Diagram
Online	test	svasic	Platform Executor	nodered-service:1880	PlatformExecutor	DETERMINISM	RUNNING	Go to Diagram
Offline	svTest	svasic	-	-	-	-	-	Go to Diagram
Offline	ss	svasic	-	-	-	-	-	Go to Diagram
Online	Skid P-205 Aux Pump Control	kopi	Platform Executor	nodered-service:1880	PlatformExecutor	DETERMINISM	RUNNING	Go to Diagram
Online	Skid P-205 BP Valve control	kopi	Platform Executor	nodered-service:1880	PlatformExecutor	DETERMINISM	RUNNING	Go to Diagram
Online	Skid P-205 Help Flow	kopi	Platform Executor	nodered-service:1880	PlatformExecutor	DETERMINISM	RUNNING	Go to Diagram
Offline	Mihailo TEst	Mihailo	-	-	-	-	-	Go to Diagram
Offline	dasd	sdasd	-	-	-	-	-	Go to Diagram

Showing per page 20 | Total entries: 75 | < 1 2 3 4 >

Deployment — Your Rules, Your Infrastructure

Same platform. Same features. Your choice of where it runs.

Model	What it means
Cloud (SaaS)	We host and manage — you just log in
On-Premise	Fully on your servers, your data center, your control
Hybrid	Some on-site, some cloud — best of both worlds
Edge + Cloud	Local autonomy + cloud visibility and analytics

Security — verified architecture, with regular penetration testing cycles to stay ahead of threats.

Tarkett — The Challenge

Multiple divison in factory. Multi-stage production process. SAP-connected.

Before inVIEW MES:

- Production was planned in **SAP** — but execution on the floor was disconnected
- No real-time tracking — no machinery monitoring, work orders tracking on sheets
- Material in-out and Mach Stoppage evidence manually — hours later, sometimes not at all
- Shift reports took **45 minutes** to complete manually — every single shift
- SAP always lagged reality — no one trusted the numbers

> *"Management could see what was planned. Nobody could see what actually happened."*

Tarkett Wood — SCADA + MES (since 2021)

A full Manufacturing Execution System built on the inVIEW SCADA foundation.

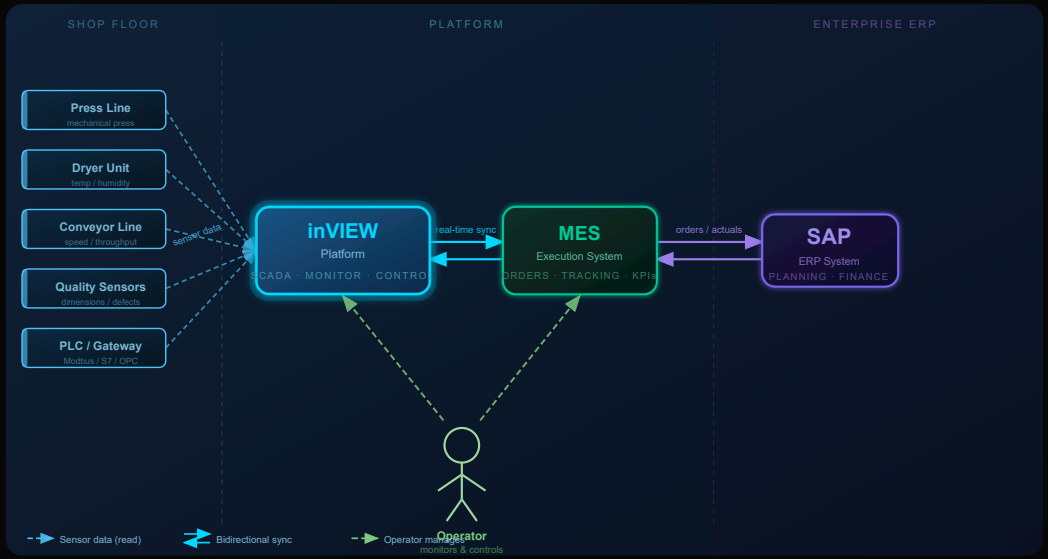
SCADA layer — real-time visibility:

- Full production floor visualization — presses, brush lines, packaging, all zones live
- Real-time machine status, cycle parameters, trend analysis
- TV screens on the shop floor — always-on production overview for operators

MES layer — planning and execution:

- **Production planning** — visual planning board, shift scheduling, line assignments
- **Work order lifecycle** — SAP order → floor screen → execution tracked step by step
- **Cycle evidence** — every press cycle recorded: duration, parameters, outcome
- **Downtime management** — stoppages auto-detected, reason codes logged, duration tracked
- **Quality & yield** — per-order data, reject tracking, product version management
- **Shift reports** — generated automatically at shift end. Zero manual input.

SAP Integration — Data Flows Both Ways



inVIEW and SAP talk to each other — automatically.

inVIEW → SAP (data flowing back to ERP):

- Actual production quantities confirmed per order
- Material consumption reported
- Downtime events and stoppage reasons
- Work order status updated in real time
- Quality deviations flagged

SAP → inVIEW (data flowing down to the floor):

- Production orders pushed directly to floor screens
- Material requirements, BOMs, product versions
- Shift schedules and planning parameters

> *"SAP always reflects what actually happened on the floor. Not because someone entered data — because the system entered it automatically."*

TARKETT WOOD — SCADA LIVE VIEW

TARKETT WOOD — DAILY PRODUCTION EVIDENCE

RADNA STANICA	A KLASA	DORADA MD	DORADA RD	ŠKART	GRAND TOTAL
PBP400 - Presa	6821.45M2	0M2	0M2	36.18M2	6857.63M2

MATERIJAL	ISPUNJENOST	PLAN A KLASE	A KLASA	DORADA MD	DORADA RD	ŠKART	GRAND TOTAL	ISPUNJENOST PRVE KLASE
940099235 - PRE TIM13-OAKOR200x2289-CT87-WAVE	36.4%	2453.28M2	893.1M2	0M2	0M2	0M2	893.1M2	100%
940257417 - PRE ALG14-OAKNA200x2289-B3-NT96	100.95%	896.07M2	904.55M2	0M2	0M2	0M2	904.55M2	100%
940317970 - PRE SAL14-OAKOR200x2289-B2-PT96	89.56%	5609.6M2	5023.8M2	0M2	0M2	36.18M2	5059.98M2	99.28%

PBP420 - Braširanje	0M2	0M2	0M2	0M2	0M2
PBP450 - Braširanje Četkanje	5936.59M2	0M2	0M2	0M2	5936.59M2
PBP455 - Digitalna stampa	0M2	0M2	0M2	0M2	0M2
PBP456 - Novi digi	0M2	0M2	0M2	0M2	0M2
PBP465 - Rucni bevel	0M2	0M2	0M2	0M2	0M2
PBP470 - Novi homag	3549.43M2	0M2	0M2	0M2	3549.43M2
PBP480 - BKS	0M2	0M2	0M2	0M2	0M2
PBP490 - Uljenje	0M2	0M2	0M2	0M2	0M2
PBP500 - Lakiranje	2406.06M2	0M2	0M2	0M2	2406.06M2
PBP550 - Rucna dorada	0M2	0M2	0M2	0M2	0M2
PBP600 - Stari homage	3523.13M2	0M2	0M2	0M2	3523.13M2

Validacija ciklusa

Vreme završetka radnog ciklusa:

15. 2. 2026., 22:00:00

Trajanje ciklusa	480.00 min	14:00 - 22:00	
Vreme rada	66.00 min	Brzina proizvodnje	4235 m^2/8h
		A klasa	582.32 m^2
		RD	0 m^2
		MD	0 m^2
		Škart	0 m^2
		Ukupno	582.32 m^2
Zastoji	0.00 min		
Mikrozastoj	**414.00 min**	**86.25 %**	

Mikrozastoji su 86.25% što je veće od 5.00%

Knjiženje | Zastoj

Evidentirana A klasa

Tarkett — Results

Wood Division (MES since 2021):

- Production planning fully digitalized and connected to SAP
- Every work order tracked from SAP creation to floor completion
- Every press cycle recorded — full production traceability
- Downtime automatically detected and categorized
- **Shift reports automated** — 45 minutes of manual work → 0
- Live quality data per order, per shift, per product version

PVC Division *(running since 2016 — mentioned for reference):*

- Full SCADA monitoring and control across all production zones
- OEE tracking, command system, SAP-integrated production orders

> *"They started with one line in 2016. Today they run two full divisions on inVIEW. That's 10 years of continuous expansion — the real proof isn't the sale, it's that they keep growing it."*

Thank You

inVIEW IIoT Platform — INIT Technologies

The platform that connects your machines, your people, and your data.

Next step: A short technical workshop at your site — we look at your current setup and define what a pilot could look like.

Robert Sabo | INIT Technologies
robert@industrial-it.software | inview.software

> "*You already have the data. Let's make it work for you.*"